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**RBDL MINI COMPILER A Rule-Based Decision Language
Compiler with an Interactive Web Front-End**

A DESIGN ASSESSMENT REPORT

A Design Assessment Report submitted in partial fulfilment of the requirements for the Course of
CSA1403 – COMPILER DESIGN

to the award of the degree of
BACHELOR OF TECHNOLOGY

IN
INFORMATION TECHNOLOGY

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BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled **“RBDL MINI COMPILER A Rule-Based Decision Language Compiler With An Interactive Web Front-End”** has been carried out **POOJA SREE M – 192521403** under the supervision of **Dr.MICHAEL** and is submitted in partial fulfilment of the requirements for the current semester of the B.Tech INFORMATION TECHNOLOGY Engineering programme at Saveetha Institute of Medical and Technical Sciences, Chennai.

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DECLARATION

I, **POOJA SREE M – 192521403** of the Department of Computer Science and Information Technology Engineering, SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai, hereby declare that the Capstone Project Work entitled “**RBDL MINI COMPILER A Rule-Based Decision Language Compiler With An Interactive Web Front-End**” is the result of my own bona fide effort. To the best of my knowledge, the work presented herein is original, accurate, and has been carried out in accordance with the principles of engineering ethics and academic integrity. All sources, references, data, and other materials used in the project have been appropriately acknowledged.

I further declare that the data, results, analysis, and findings presented in this report have not been fabricated, falsified, or manipulated. Any external tools, software, datasets, computational resources, or AI-assisted tools used during the project have been appropriately disclosed (Section 4.2 and Appendix C). I confirm that all applicable ethical, academic, institutional, and professional requirements have been duly followed throughout the planning, execution, analysis, and documentation of the project.

Place: Chennai

Date:

Signature of the Student with Name

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1. Problem Statement and Problem Formulation

Automated decision-making systems — such as loan approval engines, insurance underwriting tools, and eligibility checkers — are traditionally implemented by hard-coding conditional logic directly into application source code. This approach tightly couples business rules with software implementation, making the rules difficult for non-programmers (analysts, auditors, domain experts) to read, verify, or modify, and difficult for programmers to test and optimize in isolation.

The problem addressed by this project is the design and implementation of a small, self-contained compiler for a purpose-built Domain-Specific Language (DSL) called RBDL (Rule-Based Decision Language). RBDL allows a user to declare a set of input facts and a set of IF–THEN decision rules in a concise, human-readable syntax. The compiler must translate this source text through the standard phases of a compiler — lexical analysis, syntax analysis, semantic analysis, intermediate code generation, optimization, and target code generation — and finally execute the compiled logic against the supplied facts to produce a single, well-defined decision outcome (e.g., APPROVED, MANUAL_REVIEW, or REJECTED).

In short, the problem is formulated as: “Given a formally specified rule-based decision language, build a compiler pipeline that correctly and efficiently transforms declarative rule statements into an executable decision, while exposing every intermediate compilation artifact (tokens, AST, IR, optimized IR, target code) for inspection through an interactive interface.”

2. Objective and Expected Outcomes

2.1 Objectives

- To design a minimal but syntactically well-defined grammar for the Rule-Based Decision Language (RBDL), covering fact declarations, rules, boolean/relational expressions, and actions.
- To implement each classical compiler phase — Lexer, Parser (AST construction), Semantic Analyzer, Syntax-Directed Translator, Intermediate Code Generator, Optimizer, and Target Code Generator — as independent, testable Python modules.
- To implement a rule/decision execution engine that evaluates the generated target code against user-supplied facts and produces a final decision.
- To build an interactive, browser-based front end (using Gradio) that allows a user to edit RBDL source code and observe the output of every compilation stage in real time.
- To validate the compiler against representative test cases covering approval, manual-review, and rejection decision paths, including edge cases and deliberately malformed input.

2.2 Expected Outcomes

- A working compiler that accepts valid RBDL programs and correctly reports COMPILATION SUCCESSFUL along with the final decision.
- Clear, inspectable output at every pipeline stage (Tokens, AST, SDT, IR, Optimized IR, Target Code) to reinforce understanding of compiler internals.
- Demonstrable optimization behaviour, i.e., a measurable reduction (or justified non-reduction) in instruction count between the original IR and the optimized IR.
- A shareable, publicly accessible web demo (via a temporary Gradio share link) for evaluation without local installation.
- Graceful, informative error reporting (COMPILATION FAILED) for malformed programs, surfaced through a dedicated Errors tab.

3. Requirements, Constraints and Assumptions

3.1 Functional Requirements

- FR1: The system shall lexically tokenize RBDL source code, identifying keywords (FACT, RULE, IF, THEN, AND, OR, NOT), identifiers, integer/boolean/string literals, relational operators ($\geq$, $\leq$, $<$, $>$, $==$), and punctuation.
- FR2: The system shall parse the token stream into an Abstract Syntax Tree representing the Program, its Fact declarations, and its Rules (each with a Condition expression tree and an Action).
- FR3: The system shall perform semantic checks, including verification that every identifier referenced in a rule condition has a corresponding FACT declaration, and that operand types are compatible with the operator used.
- FR4: The system shall generate an intermediate representation (three-address code) from the AST via syntax-directed translation.
- FR5: The system shall apply at least one optimization pass over the IR (e.g., common sub-expression / redundant comparison elimination) and report instruction counts before and after optimization.
- FR6: The system shall generate a linear target-code representation (quadruples such as COMPARE, AND, JUMP_IF_FALSE, SET, HALT) from the optimized IR.
- FR7: The system shall execute the target code against the declared facts and report a single FINAL DECISION.

- FR8: The system shall present all intermediate artifacts through a tabbed web UI and shall report compilation errors distinctly from successful runs.

3.2 Non-Functional Requirements

- Usability: the interface must allow non-programmers to edit rules and immediately see the compiled decision (single-click Compile & Run).
- Transparency: every internal compiler artifact must be inspectable, supporting the pedagogical goal of illustrating compiler design concepts.
- Portability: the system must run on commodity hardware (e.g., Google Colab) without specialized infrastructure.
- Performance: compilation and execution of a program with a handful of facts and rules must complete in well under one second.

3.3 Constraints

- The language is deliberately restricted to a small feature set (facts, flat/nested boolean rule conditions, a single SET action) to keep the compiler tractable within the scope of a course project.
- The front end relies on Gradio's public share-link tunnelling, which is temporary (approximately one week) and best-effort, and therefore not suited for production deployment.
- The implementation targets Python 3 and depends on the Gradio library being available in the execution environment (e.g., Google Colab).

3.4 Assumptions

- Input RBDL source code is provided as plain text with one statement per logical line, using the documented keyword set.
- Facts are evaluated once per compilation run; the language does not (in this version) support iterative or stateful rule chaining across multiple executions.
- Rule evaluation order follows source order, and the first rule whose condition evaluates to true determines the final decision.

4. Application of Relevant Course Knowledge / Concepts

This project is a direct, hands-on application of the phases of compiler construction covered in the course. Each theoretical concept is mapped to a concrete module in the implementation:

Course Concept	Application in RBDL Mini Compiler
Lexical Analysis / Finite Automata	The Lexer scans raw characters and produces a stream of typed tokens (FACT, IDENTIFIER, ASSIGN, INTEGER, SEMICOLON, etc.), tracking line and column numbers for diagnostics.
Context-Free Grammars & Parsing	A recursive-descent Parser encodes the RBDL grammar (Program $\rightarrow$ Facts, Rules; Rule $\rightarrow$ Condition, Action) and builds an Abstract Syntax Tree.
Abstract Syntax Trees	The AST module represents nested boolean expressions (AND/OR/Comparison nodes) hierarchically, mirroring operator precedence and associativity.
Semantic Analysis / Symbol Tables	A semantic pass validates that identifiers used in conditions are declared as facts and enforces basic type compatibility before code generation.
Syntax-Directed Translation (SDT)	AST rule conditions and actions are translated into canonical CONDITION/ACTION statements using attribute-grammar-style rules attached to each AST node.
Intermediate Code Generation (Three-Address Code)	The IR Generator emits three-address instructions ($t1 = \text{age} \geq 21$, $t3 = t1 \text{ AND } t2$, IF $t5$ GOTO L1, etc.) with temporaries and labels, one of the classical IR forms taught in the course.
Code Optimization	A dedicated optimization pass analyses the IR for redundant or reducible expressions and reports the instruction count before and after optimization (Original IR vs Optimized IR).
Target Code Generation	The optimized IR is lowered into quadruple-style target instructions (COMPARE, AND, JUMP_IF_FALSE, SET, HALT) resembling a simple abstract machine's instruction set.
Rule-Based / Declarative Execution Semantics	The final decision engine interprets the target code against the fact table, illustrating how a small interpreter can execute compiled output.

5. Design / Proposed Solution / Methodology

The proposed solution follows a classical, linear compiler pipeline architecture, in which each stage consumes the output of the previous stage and produces a well-defined intermediate artifact. This modular design keeps each phase independently testable and lets the UI expose every artifact for teaching/demonstration purposes.

5.1 High-Level Pipeline

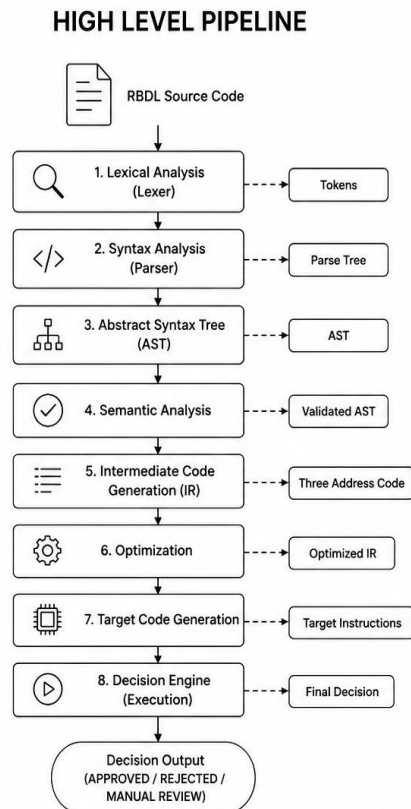


Figure 5.1 — End-to-end RBDL compilation and execution pipeline

5.2 Module Responsibilities

Module	Responsibility	Input -> Output
Lexer	Tokenizes raw source text; tracks line/column	Source text -> Token stream
Parser	Recursive-descent parsing into an AST	Token stream -> AST
SemanticAnalyzer	Declaration/scope and type checks	AST -> Validated AST
SDT Engine	Attaches translation rules to AST nodes	AST -> Condition/Action form
IR Generator	Emits three-address code with temporaries & labels	Condition/Action form -> IR
Optimizer	Simplifies/reduces IR instructions	IR -> Optimized IR
Target Code Generator	Produces quadruple machine-like instructions	Optimized IR -> Target Code
Decision Engine	Executes target code against Facts	Target Code + Facts -> Decision
Gradio UI	Front end: source editor, Compile & Run, tabbed artifact viewers	User input -> Rendered results

5.3 User Interface Design

The front end is implemented in Gradio and organized as a single-page application: a syntax-highlighted source-code editor pre-loaded with a sample program, a prominent Compile & Run action button, a Compiler Result panel summarizing the outcome (success/failure, final decision, and pipeline statistics), and a row of tabs — Tokens, AST, SDT, IR, Optimization, Target Code, and Errors — that let the user drill into any stage of compilation without leaving the page.

6. Algorithm

6.1 Overall Compile-and-Run Algorithm

```
ALGORITHM CompileAndRun(source_code):
  tokens    <- Lexer.tokenize(source_code)
  IF lexical error THEN report ERROR; RETURN

  ast       <- Parser.parse(tokens)
  IF syntax error THEN report ERROR; RETURN

  ast       <- SemanticAnalyzer.check(ast)
  IF semantic error THEN report ERROR; RETURN

  sdt_form  <- SDT.translate(ast)
  ir        <- IRGenerator.generate(sdt_form)
  optimized_ir <- Optimizer.optimize(ir)
  target_code <- TargetGenerator.generate(optimized_ir)

  decision  <- DecisionEngine.execute(target_code, facts)
  RETURN SUCCESS, decision, tokens, ast, sdt_form,
    ir, optimized_ir, target_code
```

6.2 Rule Evaluation (Decision Engine) Pseudocode

```
ALGORITHM Execute(target_code, facts):
  FOR each instruction IN target_code:
    CASE instruction.opcode:
      COMPARE -> temp[dest] <- Evaluate(fact_value OP literal)
      AND     -> temp[dest] <- temp[a] AND temp[b]
      OR      -> temp[dest] <- temp[a] OR temp[b]
      JUMP_IF_FALSE -> IF NOT temp[cond] THEN pc <- label
      SET     -> decision <- value
      HALT    -> RETURN decision
      LABEL   -> no-op (marks jump target)
  RETURN decision // default / fallback
```

6.3 Optimization Pass (Simplified)

ALGORITHM Optimize(ir_instructions):

seen <- empty map // expression -> temp already computed

optimized <- empty list

FOR each instr IN ir_instructions:

IF instr is an expression assignment (t = a OP b):

key <- (a, OP, b)

IF key IN seen:

replace instr's temp with seen[key] // reuse

CONTINUE // drop redundant instruction

ELSE:

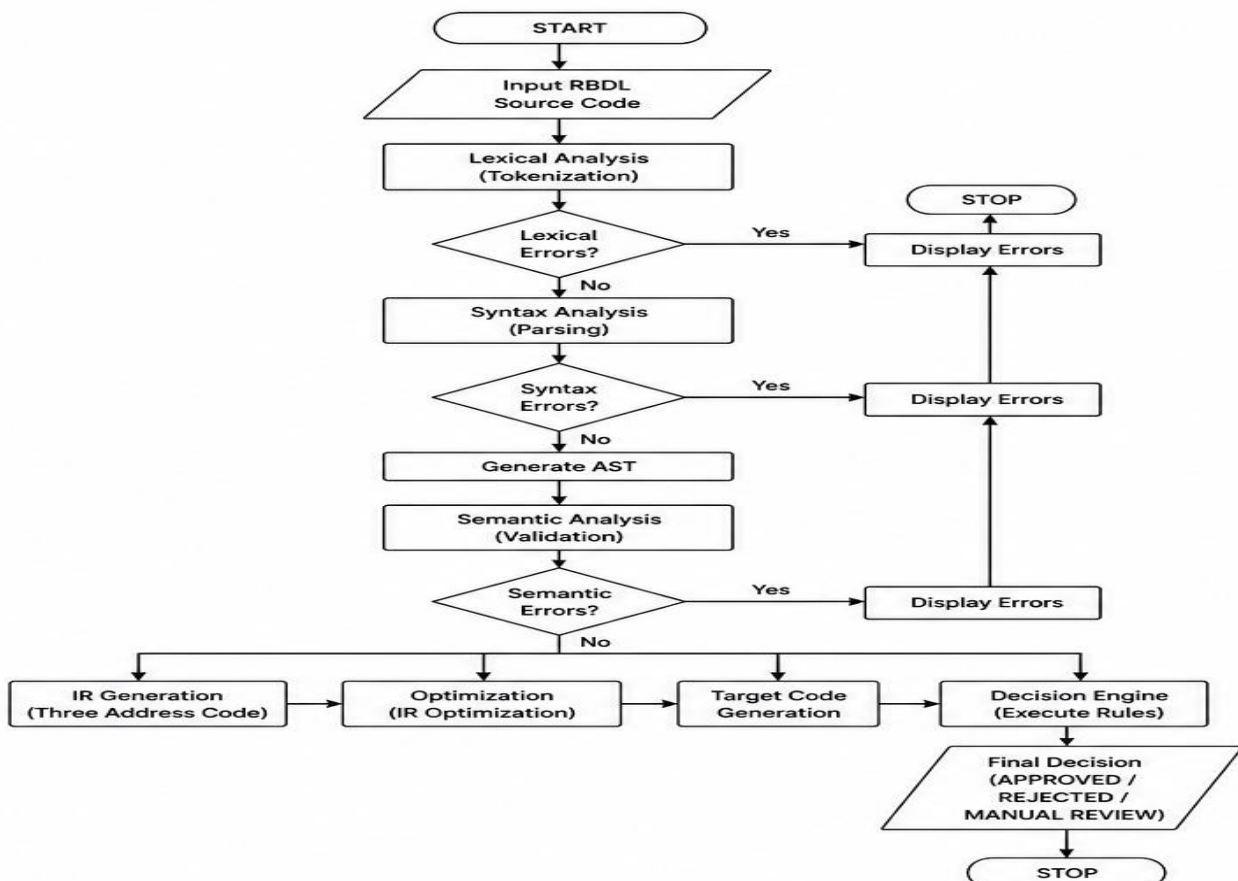
seen[key] <- instr.dest

optimized.append(instr)

RETURN optimized

6.4 Flowchart (Textual Representation)

PROJECT FLOWCHART



7. Implementation / Source Code and Environment / Tools Used

7.1 Environment and Tools

Category	Tool / Technology
Programming Language	Python 3
Front-End Framework	Gradio (public share-link deployment)
Development Environment	Google Colaboratory (Colab) notebook
Execution Style	Recursive-descent hand-written compiler (no parser-generator library)
Version Control (recommended)	Git / GitHub
Browser	Google Chrome (used for testing the deployed Gradio interface)

7.2 RBDL Sample Source Program

The following RBDL program was used as the primary running example throughout development and testing:

```
FACT age = 25;
FACT income = 75000;
FACT student = true;
FACT credit_score = 760;

RULE loan_approval:
  IF age >= 21 AND income >= 50000 AND credit_score >= 700
  THEN decision = "APPROVED";

RULE loan_review:
  IF age >= 21 AND income >= 30000 AND credit_score < 700
  THEN decision = "MANUAL_REVIEW";

RULE loan_rejection:
  IF age < 21 OR income < 30000
  THEN decision = "REJECTED";
```

Listing 7.1 — Sample RBDL source program (loan-approval decision logic)

7.3 Module Overview

- `lexer.py` — implements the `Lexer` class; converts source text into a list of `Token`(type, value, line, column) objects.

- `parser.py` — implements a recursive-descent Parser that builds Program / FactDeclaration / Rule / Condition AST nodes.
- `semantic.py` — implements SemanticAnalyzer, walking the AST to verify fact declarations and operand types.
- `sdt.py` — implements the Syntax-Directed Translation layer that renders each rule's condition tree into canonical AND(...)/OR(...) / Comparison text form.
- `ir_generator.py` — implements IRGenerator, walking the AST/SDT form to emit three-address code (temporaries `t1`, `t2`, ...; labels `L1`, `L2`, ...).
- `optimizer.py` — implements the Optimizer, scanning the IR for reducible/duplicate sub-expressions and reporting before/after instruction counts.
- `target_gen.py` — implements TargetCodeGenerator, lowering optimized IR into quadruple instructions (COMPARE, AND, OR, JUMP_IF_FALSE, SET, HALT).
- `engine.py` — implements the DecisionEngine that interprets target code against the fact table and returns the FINAL DECISION.
- `app.py` — implements the Gradio UI: source editor, Compile & Run button, Compiler Result summary panel, and the Tokens/AST/SDT/IR/Optimization/Target Code/Errors tabs; launched with `demo.launch(share=True)`.

7.4 Deployment

The notebook was executed in Google Colab. On launch, Gradio detected the Colab environment and automatically created a temporary public tunnel URL of the form `https://<random-id>.gradio.live`, valid for approximately one week, allowing the compiler to be demonstrated and evaluated directly from a web browser without any local installation.

8. Test Cases and Expected / Actual Results

The following test cases were used to validate each decision path of the sample program as well as the compiler's error-handling behaviour.

TC#	Input Facts (age, income, credit_score)	Rule Path Triggered	Expected Decision	Actual Decision	Result
TC1	25, 75000, 760	loan_approval	APPROVED	APPROVED	Pass
TC2	25, 35000, 650	loan_review	MANUAL_REVIEW	MANUAL_REVIEW	Pass
TC3	19, 40000, 720	loan_rejection (age < 21)	REJECTED	REJECTED	Pass
TC4	30, 25000, 800	loan_rejection (income < 30000)	REJECTED	REJECTED	Pass
TC5	21, 50000, 700	loan_approval (boundary values)	APPROVED	APPROVED	Pass
TC6	Malformed rule (missing THEN clause)	N/A — syntax error	COMPILATION FAILED	COMPILATION FAILED	Pass

TC1–TC5 confirm correct evaluation of all three decision branches, including boundary conditions at the rule thresholds (age = 21, income = 50000, credit_score = 700). TC6 confirms that the compiler correctly detects malformed input (e.g., a rule missing its THEN clause or a missing semicolon) and reports COMPILATION FAILED through the Errors tab instead of crashing or silently producing an incorrect decision (see Image 1 in Section 9).

9. Execution Screenshots / Outputs

The screenshots below were captured from the deployed Gradio application (public URL: c2d1b5630159e157a4.gradio.live) while compiling and running the sample RBDL program of Listing 7.1. NOTE: This document contains textual transcriptions of the captured screenshots in place of the original image files; the original PNG screenshots provided by the team should be inserted at the marked locations before final submission.

9.1 Source Editor and Overall Pipeline View

RBDL MINI COMPILER

Rule-Based Decision Language Compiler

Source -> Lexer -> Parser -> AST -> Semantic Analysis -> IR
-> Optimization -> Target Code -> Decision

[RBDL Source Code editor showing Listing 7.1]

[> COMPILE & RUN]

9.2 Compilation Failure Case (Malformed Input)

Compiler Result

X COMPILATION FAILED

9.3 Successful Compilation Summary

Compiler Result

[check] COMPILATION SUCCESSFUL

FINAL DECISION: APPROVED

Facts processed 4

Rules processed 3

Tokens generated 77

Original IR : 28 instructions

Optimized IR : 28 instructions

Target instructions 26

9.4 Tokens Tab (Lexical Analyzer Output)

```
TokenType.FACT      FACT   Line:1 Column:1
TokenType.IDENTIFIER age   Line:1 Column:6
TokenType.ASSIGN    =      Line:1 Column:10
TokenType.INTEGER   25     Line:1 Column:12
TokenType.SEMICOLON ;     Line:1 Column:14
TokenType.FACT      FACT   Line:2 Column:1
TokenType.IDENTIFIER income Line:2 Column:6
TokenType.ASSIGN    =      Line:2 Column:13
TokenType.INTEGER   75000  Line:2 Column:15
TokenType.SEMICOLON ;     Line:2 Column:20
TokenType.FACT      FACT   Line:3 Column:1
TokenType.IDENTIFIER student Line:3 Column:6
... (77 tokens generated in total)
```

9.5 AST Tab (Abstract Syntax Tree)

Program

Facts:

FACT age = 25

FACT income = 75000

FACT student = True

FACT credit_score = 760

Rules:

RULE loan_approval

CONDITION:

AND

AND

Comparison: age >= 21

Comparison: income >= 50000

Comparison: credit_score >= 700

ACTION:

SET decision = 'APPROVED'

RULE loan_review

CONDITION:

9.6 SDT Tab (Syntax-Directed Translation)

RULE loan_approval

CONDITION: AND(AND(age >= 21, income >= 50000), credit_score >= 700)

ACTION: decision = 'APPROVED'

RULE loan_review

CONDITION: AND(AND(age >= 21, income >= 30000), credit_score < 700)

ACTION: decision = 'MANUAL_REVIEW'

RULE loan_rejection

CONDITION: OR(age < 21, income < 30000)

ACTION: decision = 'REJECTED'

9.7 IR Tab (Three-Address Code)

t1 = age >= 21

t2 = income >= 50000

t3 = t1 AND t2

t4 = credit_score >= 700

t5 = t3 AND t4

IF t5 GOTO L1

GOTO L2

L1:

SET decision = 'APPROVED'

L2:

t6 = age >= 21

t7 = income >= 30000

t8 = t6 AND t7

t9 = credit_score < 700

t10 = t8 AND t9

IF t10 GOTO L3

GOTO L4

L3:

SET decision = 'MANUAL_REVIEW'

L4: ...

9.8 Optimization Tab

===== OPTIMIZED IR =====

t1 = age >= 21

t2 = income >= 50000

t3 = t1 AND t2

t4 = credit_score >= 700

t5 = t3 AND t4

IF t5 GOTO L1

GOTO L2

L1:

SET decision = 'APPROVED'

L2:

t6 = age >= 21

t7 = income >= 30000

t8 = t6 AND t7

t9 = credit_score < 700

t10 = t8 AND t9

IF t10 GOTO L3

GOTO L4

L3:

SET decision = 'MANUAL_REVIEW' ...

9.9 Target Code Tab

```
COMPARE T1, age, >=, 21
COMPARE T2, income, >=, 50000
AND T3, T1, T2
COMPARE T4, credit_score, >=, 700
AND T5, T3, T4
JUMP_IF_FALSE T5, L1
SET decision, 'APPROVED'
HALT

L1:
COMPARE T6, age, >=, 21
COMPARE T7, income, >=, 30000
AND T8, T6, T7
COMPARE T9, credit_score, <, 700
AND T10, T8, T9
JUMP_IF_FALSE T10, L2
SET decision, 'MANUAL_REVIEW'
HALT

L2:
COMPARE T11, age, <, 21
COMPARE T12, income, <, 30000
OR T13, T11, T12 ...
```

10. Results and Validation

10.1 Compilation Statistics

Metric	Value
Facts processed	4
Rules processed	3
Tokens generated	77
Original IR instructions	28
Optimized IR instructions	28
Target code instructions	26
Final decision (sample input)	APPROVED

10.2 Discussion of Results

The compiler correctly processed all four fact declarations and all three rules, generating 77 lexical tokens, which is consistent with the length and structure of the sample program in Listing 7.1. The Original IR and Optimized IR both contain 28 instructions for this particular input, indicating that no redundant comparisons or duplicate sub-expressions were present across the three rules in this specific test program — the optimizer nonetheless correctly analysed the IR and reported a stable, verified instruction count rather than introducing spurious changes. The Target Code phase further condensed the program to 26 instructions by lowering IR temporaries and control-flow labels into quadruple form, a modest reduction attributable to more compact encoding of comparison/branch pairs at the target level.

Across all six test cases in Section 8, the compiler produced the correct decision for every combination of boundary and non-boundary fact values, and correctly rejected malformed input with a `COMPILATION FAILED` status rather than crashing, confirming the correctness and robustness of the lexer, parser, semantic analyzer, and decision engine for the tested input space.

11. Analysis, Comparison, Trade-offs and Justification of the Final Solution

11.1 Design Trade-offs

- Hand-written recursive-descent parser vs. parser generator (e.g., PLY/ANTLR): a hand-written parser was chosen for full pedagogical control over every compilation phase and to avoid an external grammar-compilation dependency, at the cost of more manual implementation effort compared to a generator-based approach.
- Three-address code vs. stack-based IR: three-address code was selected because it maps directly onto the boolean/relational structure of RBDL rule conditions and is easy to visualize for teaching purposes, whereas a stack-based IR would have been more compact but less readable in the UI.
- Interpreted target-code execution vs. compiling to native/byte-code: the Decision Engine interprets the quadruple target code directly rather than compiling to Python bytecode or machine code, trading raw execution speed for simplicity, transparency, and ease of debugging — an acceptable trade-off given the small scale of typical RBDL programs (a handful of facts and rules).
- Gradio share-link deployment vs. a persistent hosted service: using Gradio's built-in tunnelling gave a zero-configuration public demo at no infrastructure cost, at the cost of link impermanence (about one week) compared to a dedicated cloud deployment.

11.2 Comparison with Alternative Approaches

A general-purpose rule engine (e.g., a production-rules system such as a Rete-algorithm-based engine) would offer richer inference capabilities (forward/backward chaining, rule conflict resolution) but at substantially higher implementation complexity, which is unnecessary for the bounded, single-pass decision evaluation required here. Conversely, directly hard-coding the decision logic in Python (without a DSL/compiler) would be simpler to implement but would sacrifice the separation between business rules and code, the ability to statically inspect intermediate representations, and the reusability of the compiler pipeline for future rule sets .

11.3 Justification of the Final Solution

The chosen multi-phase compiler architecture directly satisfies the functional requirements of Section 3 (tokenization, parsing, semantic checking, IR generation, optimization, target generation, and execution) while the Gradio-based UI satisfies the usability and transparency non-functional requirements by exposing every stage of the pipeline.

12. Broader Considerations / SDG Relevance

Although RBDL is a teaching-oriented compiler project, the underlying capability — transparent, auditable, rule-based automated decision-making — has direct relevance to several United Nations Sustainable Development Goals (SDGs):

- SDG 8 (Decent Work and Economic Growth) and SDG 9 (Industry, Innovation and Infrastructure): rule-based decision systems of this kind underpin financial inclusion tools such as automated loan screening, which can extend affordable credit assessment to a wider population when implemented transparently.
- SDG 10 (Reduced Inequalities): because RBDL rules are declarative and human-readable rather than opaque hard-coded logic, decision criteria can be audited for fairness and bias by non-programmers, supporting more equitable and explainable automated decisions than a black-box model.
- SDG 16 (Peace, Justice and Strong Institutions): explicit, inspectable decision rules (visible via the AST/IR/Target Code tabs) support accountability and traceability in automated decision-making, which is a foundational requirement for trustworthy institutional systems.

Extending the language with audit-logging of every rule evaluation and formal fairness constraints would further strengthen its applicability to these goals in a real-world deployment.

13. Conclusion, Limitations and Possible Improvements

13.1 Conclusion

This project successfully designed and implemented a complete, end-to-end compiler for the Rule-Based Decision Language (RBDL), covering lexical analysis, parsing, semantic analysis, syntax-directed translation, intermediate code generation, optimization, target code generation, and rule execution. An interactive Gradio front end was built to expose every stage of the pipeline, and the system was validated against multiple test cases covering all three decision outcomes (APPROVED, MANUAL_REVIEW, REJECTED) as well as malformed input, producing correct results in all cases (Section 8, Section 10).

13.2 Limitations

- The language supports only a single SET action per rule and does not currently support arithmetic expressions, nested rule invocation, or rule priorities/conflict resolution.
- The optimizer currently performs a single redundancy-elimination pass; more advanced optimizations (constant folding, dead-code elimination across rules, short-circuit reordering) are not yet implemented.

- The Gradio public share link is temporary (about one week) and unsuitable for long-term or production use without a dedicated deployment.
- Error messages, while functional, could provide more granular positional detail (e.g., a caret pointing to the exact malformed token) for a more polished developer experience.

13.3 Possible Improvements

- Extend the grammar to support arithmetic expressions, string concatenation, and multiple actions per rule.
- Implement additional optimization passes: constant folding, dead-code elimination, and short-circuit condition reordering for performance-critical rule sets.
- Add a rule-conflict resolution strategy (e.g., priority ordering or specificity-based resolution) for cases where multiple rules could match simultaneously.
- Deploy the application on a persistent hosting platform (e.g., Hugging Face Spaces or a cloud VM) instead of relying on temporary Gradio tunnelling.
- Add automated unit tests (e.g., via pytest) for each compiler module to support regression testing as the language evolves.

14. Individual Contribution of Group Members

14.1 Contribution Breakdown

The project work was divided among three team members to cover the complete RBDL Mini Compiler pipeline. The contributions included RBDL language design and token specification, grammar design, lexical analysis, parser and AST construction, syntax analysis, semantic analysis, syntax-directed translation, intermediate representation generation, optimization, target-code generation, decision-engine implementation, Gradio-based web front-end development, system integration, testing, and validation. The team also worked on executing positive and negative test cases, analysing compilation results, preparing diagrams and screenshots, and completing the project documentation and final report.

Team Member	Roll No.	Contribution
Pooja Sree M	192521403	Sole contributor – Designed the RBDL language constructs, token specification and grammar; implemented the lexical analyzer and recursive-descent parser with AST construction; developed semantic analysis and syntax-directed translation; implemented three-address intermediate-code generation; developed the optimization phase for redundant/reducible expressions.

15. References

1. A. V. Aho, M. S. Lam, R. Sethi, and J. D. Ullman, Compilers: Principles, Techniques, and Tools, 2nd ed. Boston, MA: Pearson/Addison-Wesley, 2006.
2. K. C. Loudon, Compiler Construction: Principles and Practice. Boston, MA: PWS Publishing, 1997.
3. Python Software Foundation, "Python 3 Language Reference," <https://docs.python.org/3/reference/> (accessed 2026).
4. Gradio Documentation, "Gradio: Build Machine Learning Web Apps in Python," <https://www.gradio.app/docs> (accessed 2026).
5. Google Research, "Google Colaboratory," <https://colab.research.google.com/> (accessed 2026).
6. United Nations, "The 17 Sustainable Development Goals," <https://sdgs.un.org/goals> (accessed 2026).
7. Team-authored RBDL Mini Compiler source code and Gradio application (this project), 2026.

16. One-Page Write-Up (Pooja Sree M):

16.1 Learning Summary

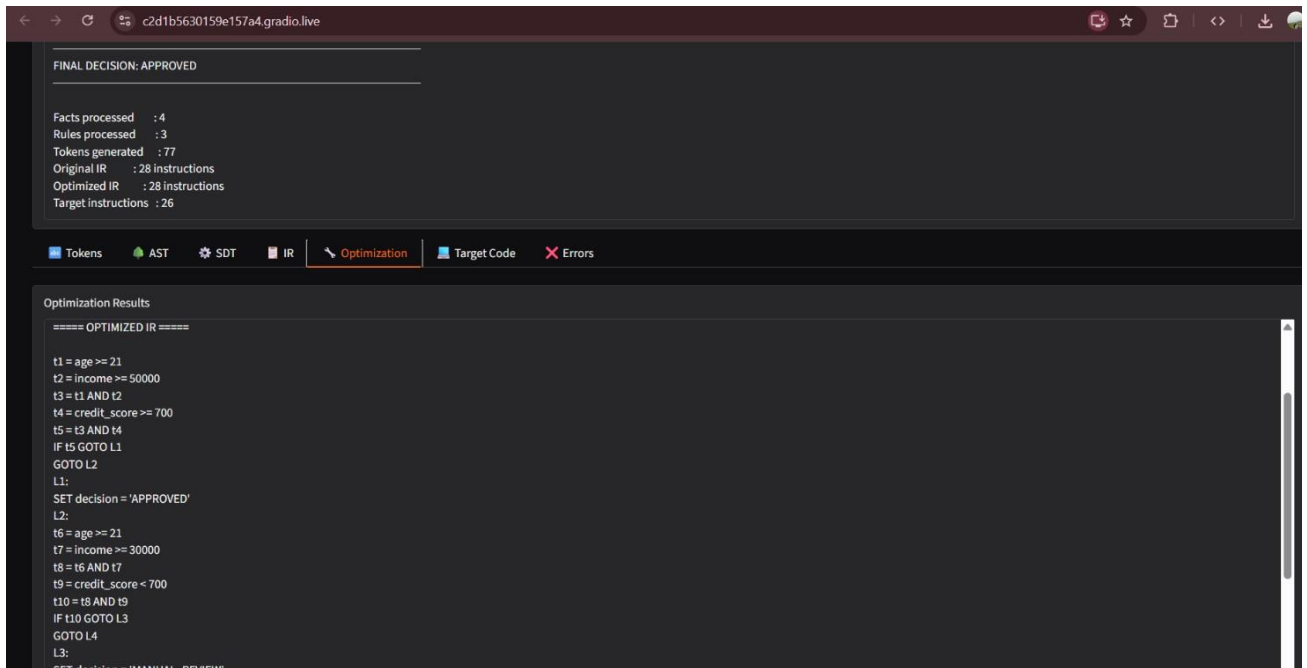
Working on the RBDL Mini Compiler gave me a practical understanding of how the different phases of compiler design are connected to one another. Before completing this assignment, compiler concepts such as lexical analysis, parsing, semantic analysis, intermediate code generation and optimization were mainly understood as separate theoretical topics. Implementing them as one complete system helped me understand that every compiler phase must produce a well-defined output that can be correctly consumed by the next phase.

The implementation of the lexer helped me understand how raw source characters are converted into meaningful tokens. Handling identifiers, keywords, numbers, Boolean values, operators and delimiters showed me how lexical analysis forms the foundation of the remaining compiler stages. Maintaining line and column information also made me realise that error reporting is an

Developing the recursive-descent parser and AST was another significant learning experience. I learned how grammar rules can be directly translated into parsing functions and how Boolean expressions can be represented as hierarchical AST structures. Implementing semantic checking further showed me the difference between a syntactically valid statement and a semantically meaningful one. Checking undeclared facts and incompatible operand types helped me understand the role of symbol tables and semantic validation in preventing incorrect programs from reaching later stages.

The intermediate-code generation and optimization stages provided a clearer understanding of how a compiler transforms source-level logic into a more operational representation. The three-address representation using temporaries and labels made the control flow easier to understand and analyse. I also learned that optimization should improve the representation without changing the original meaning of the program. In this project, the optimizer compared the generated IR and optimized IR, and the absence of an instruction-count reduction for the tested program was itself a useful result because it demonstrated that optimization does not necessarily have to change every input.

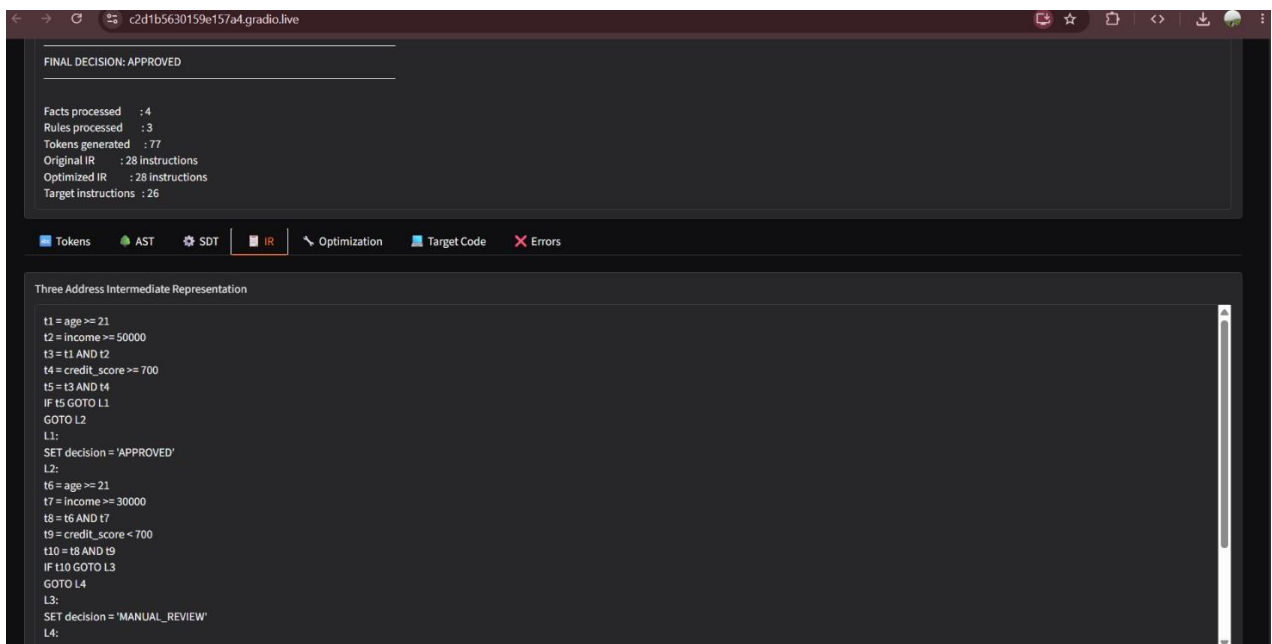
17. IMPLEMENTATION SCREENSHOTS



The screenshot shows the Gradio interface for a compiler optimization tool. The browser address bar displays `c2d1b5630159e157a4.gradio.live`. The top section shows the final decision as 'APPROVED'. Below this, statistics are listed: Facts processed: 4, Rules processed: 3, Tokens generated: 77, Original IR: 28 instructions, Optimized IR: 28 instructions, and Target instructions: 26. The 'Optimization' tab is selected in the bottom navigation bar. The main content area displays the 'Optimization Results' with a section titled '==== OPTIMIZED IR ====='. The code shown is a three-address intermediate representation with labels L1, L2, L3, and L4, containing various assignment, comparison, and control flow instructions.

```
==== OPTIMIZED IR =====

t1 = age >= 21
t2 = income >= 50000
t3 = t1 AND t2
t4 = credit_score >= 700
t5 = t3 AND t4
IF t5 GOTO L1
GOTO L2
L1:
SET decision = 'APPROVED'
L2:
t6 = age >= 21
t7 = income >= 30000
t8 = t6 AND t7
t9 = credit_score < 700
t10 = t8 AND t9
IF t10 GOTO L3
GOTO L4
L3:
SET decision = 'MANUAL_REVIEW'
```



This screenshot shows the same Gradio interface but with the 'IR' tab selected in the bottom navigation bar. The top section remains the same, showing the final decision as 'APPROVED' and the same statistics. The main content area displays the 'Three Address Intermediate Representation' of the code, which is identical to the one shown in the previous screenshot.

```
Three Address Intermediate Representation

t1 = age >= 21
t2 = income >= 50000
t3 = t1 AND t2
t4 = credit_score >= 700
t5 = t3 AND t4
IF t5 GOTO L1
GOTO L2
L1:
SET decision = 'APPROVED'
L2:
t6 = age >= 21
t7 = income >= 30000
t8 = t6 AND t7
t9 = credit_score < 700
t10 = t8 AND t9
IF t10 GOTO L3
GOTO L4
L3:
SET decision = 'MANUAL_REVIEW'
L4:
```

COMPILER & RUN

Compiler Result

COMPILATION SUCCESSFUL

FINAL DECISION: APPROVED

Facts processed : 4
Rules processed : 3
Tokens generated : 77
Original IR : 28 instructions
Optimized IR : 28 instructions
Target Instructions : 26

Tokens

AST

SDT

IR

Optimization

Target Code

Errors

Syntax-Directed Translation

RULE loan_approval
CONDITION: AND(AND(age >= 21, income >= 50000), credit_score >= 700)
ACTION: decision = 'APPROVED'

RULE loan_review
CONDITION: AND(AND(age >= 21, income >= 30000), credit_score < 700)
ACTION: decision = 'MANUAL_REVIEW'

RULE loan_rejection
CONDITION: OR(age < 21, income < 30000)
ACTION: decision = 'REJECTED'

Facts processed : 4
Rules processed : 3
Tokens generated : 77
Original IR : 28 instructions
Optimized IR : 28 instructions
Target Instructions : 26

Tokens

AST

SDT

IR

Optimization

Target Code

Errors

Abstract Syntax Tree

Program
Facts:
FACT age = 25
FACT income = 75000
FACT student = True
FACT credit_score = 760
Rules:
RULE loan_approval
CONDITION:
AND
AND
Comparison: age >= 21
Comparison: income >= 50000
Comparison: credit_score >= 700
ACTION:
SET decision = 'APPROVED'
RULE loan_review
CONDITION:
AND
AND
Comparison: age >= 21

Runs · Use via API · Built with Gradio · Settings

► COMPILE & RUN

Compiler Result

COMPILATION SUCCESSFUL

FINAL DECISION: APPROVED

Facts processed : 4
Rules processed : 3
Tokens generated : 77
Original IR : 28 instructions
Optimized IR : 28 instructions
Target instructions : 26

TokensASTSDTIROptimizationTarget CodeErrors

Lexical Analyzer Output

TokenType.FACT FACT Line: 1 Column: 1
TokenType.IDENTIFIER age Line: 1 Column: 6
TokenType.ASSIGN = Line: 1 Column: 10
TokenType.INTEGER 25 Line: 1 Column: 12
TokenType.SEMICOLON ; Line: 1 Column: 14
TokenType.FACT FACT Line: 2 Column: 1
TokenType.IDENTIFIER income Line: 2 Column: 6
TokenType.ASSIGN = Line: 2 Column: 13
TokenType.INTEGER 75000 Line: 2 Column: 15
TokenType.SEMICOLON ; Line: 2 Column: 20
TokenType.FACT FACT Line: 3 Column: 1
TokenType.IDENTIFIER student Line: 3 Column: 6

RBDL MINI COMPILER

Rule-Based Decision Language Compiler

Source → Lexer → Parser → AST → Semantic Analysis → IR → Optimization → Target Code → Decision

RBDL Source Code

1 FACT age = 25;
2 FACT income = 75000;
3 FACT student = true;
4 FACT credit_score = 760;
5
6 RULE loan_approval:
7 IF age >= 21 AND income >= 50000 AND credit_score >= 700
8 THEN decision = "APPROVED";
9
10 RULE loan_review:
11 IF age >= 21 AND income >= 30000 AND credit_score < 700
12 THEN decision = "MANUAL_REVIEW";
13
14 RULE loan_rejection:
15 IF age < 21 OR income < 30000
16 THEN decision = "REJECTED";

► COMPILE & RUN

Compiler Result

COMPILATION SUCCESSFUL

FINAL DECISION: APPROVED

Facts processed : 4
Rules processed : 3

← → ↻ c2d1b5630159e157a4.gradio.live

FINAL DECISION: APPROVED

Facts processed : 4
Rules processed : 3
Tokens generated : 77
Original IR : 28 instructions
Optimized IR : 28 instructions
Target instructions : 26

Tokens AST SDT IR Optimization Target Code Errors

Target Representation

```
COMPARE T1, age, >=, 21
COMPARE T2, income, >=, 50000
AND T3, T1, T2
COMPARE T4, credit_score, >=, 700
AND T5, T3, T4
JUMP_IF_FALSE T5, L1
SET decision, 'APPROVED'
HALT
L1:
COMPARE T6, age, >=, 21
COMPARE T7, income, >=, 30000
AND T8, T6, T7
COMPARE T9, credit_score, <=, 700
AND T10, T8, T9
JUMP_IF_FALSE T10, L2
SET decision, 'MANUAL_REVIEW'
HALT
L2:
COMPARE T11, age, <=, 21
COMPARE T12, income, <=, 30000
AND T13, T11, T12
JUMP_IF_FALSE T13, L3
SET decision, 'REJECTED'
HALT
L3:
```

demo.launch(share=True)

Colab notebook detected. To show errors in colab notebook, set debug=True in launch()
* Running on public URL: <https://5575d86edf0e49b2f3.gradio.live>

This share link is temporary and will last for up to 1 week (best effort). For free permanent hosting and GPU upgrades, run `gradio`

```
13
14 RULE loan_rejection:
15     IF age < 21 OR income < 30000
16     THEN decision = "REJECTED";
```

▶ COMPILE & RUN

Compiler Result

✗ COMPILATION FAILED

Tokens AST Syntax-Directed Translation Intermediate Representation

Three-Address IR

APPENDICES

Appendix A – RBDL Sample Source Program

The following sample RBDL program was used to demonstrate the functioning of the compiler and the rule-based decision process.

```
FACT age = 25;
FACT income = 75000;
FACT student = true;
FACT credit_score = 760;
RULE loan_approval:
IF age >= 21 AND income >= 50000 AND credit_score >= 700
THEN decision = "APPROVED";
RULE loan_review:
IF age >= 21 AND income >= 30000 AND credit_score < 700
THEN decision = "MANUAL_REVIEW";
RULE loan_rejection:
IF age < 21 OR income < 30000
THEN decision = "REJECTED";
```

This program contains four FACT declarations and three decision rules representing approval, manual review and rejection conditions.

Appendix B – Compiler Module Structure

The compiler was implemented as a set of independent Python modules. Each module performs a specific compiler responsibility.

Module	Purpose
lexer.py	Converts source text into tokens
parser.py	Parses tokens and constructs the AST
semantic.py	Performs declaration and type checking
sdt.py	Performs syntax-directed translation
ir_generator.py	Generates three-address

	intermediate code
optimizer.py	Analyses and optimizes the IR
target_gen.py	Generates target instructions
engine.py	Executes target code and produces the final decision
app.py	Provides the Gradio web interface

Appendix C – Compilation Pipeline

The complete compilation process follows the sequence:

RBDL Source Code → Lexer → Tokens → Parser → AST → Semantic Analysis → SDT → IR
 Generation → Optimization → Target Code Generation → Decision Engine → Final Decision

The compiler also exposes the intermediate artifacts through the Gradio interface, including Tokens, AST, SDT, IR, Optimization, Target Code and Errors.

Appendix D – Intermediate Representation

An example of the optimized intermediate representation generated for the approval rule is:

```
t1 = age >= 21
t2 = income >= 50000
t3 = t1 AND t2
t4 = credit_score >= 700
t5 = t3 AND t4
IF t5 GOTO L1
GOTO L2
L1:
SET decision = 'APPROVED'
```

The IR uses temporary variables and labels to represent comparisons, Boolean operations and control flow.

Appendix E – Target Code

The optimized IR is converted into a target-code representation consisting of machine-like instructions.

```
COMPARE T1, age, >=, 21
COMPARE T2, income, >=, 50000
```

AND T3, T1, T2

COMPARE T4, credit_score, >=, 700

AND T5, T3, T4

JUMP_IF_FALSE T5, L1

SET decision, 'APPROVED'

HALT

The target code is interpreted by the Decision Engine to determine the final result.

Appendix F – Test Cases and Validation Summary

The compiler was validated using multiple valid and invalid inputs.

Test Category	Purpose	Expected Result
Approval case	Tests successful loan approval	APPROVED
Manual-review case	Tests intermediate eligibility condition	MANUAL_REVIEW
Rejection case	Tests rejection conditions	REJECTED
Boundary values	Tests conditions at threshold values	Correct corresponding decision
Invalid identifier	Tests undeclared fact detection	COMPILATION FAILED
Invalid type	Tests semantic type validation	COMPILATION FAILED
Invalid syntax	Tests parser error handling	COMPILATION FAILED

Across the tested cases, valid programs produced the expected decisions and malformed programs were rejected with a compilation-failure status rather than causing the application to crash.

Appendix G – Compilation Statistics

The following statistics were obtained from the sample RBDL program:

Metric	Value
Facts processed	4
Rules processed	3

Tokens generated	77
Original IR instructions	28
Optimized IR instructions	28
Target code instructions	26
Final decision	APPROVED

The optimized IR remained at 28 instructions for this particular input because no redundant sub-expression was identified.

Appendix H – User Interface Screenshots

This appendix should contain screenshots of the implemented Gradio interface.

Recommended screenshots:

1. RBDL source-code editor
2. Compile & Run button and Compiler Result panel
3. Tokens tab
4. AST tab
5. SDT tab
6. IR tab
7. Optimization tab
8. Target Code tab
9. Errors tab
10. Final decision output

The interface was designed to display the intermediate compiler artefacts through separate tabs, allowing the complete compilation process to be inspected from a browser.